Systematic Strategy - Final Report

2026-09-05 · 93-month daily backtest, 2019-01 to 2026-09 - all results net of costs unless stated - gross exposure capped at 100% throughout

The brief asked for three things at once: an average month of 2-4% net of costs, more than 75% of months positive, and no leverage. The engine shipped here clears the second bar and misses the first. It is positive in 70 of 93 months (75.3%) and averages 1.64% per month, about 1.2x short of the 2% floor. The third condition it obeys to the decimal: gross exposure peaks at exactly 100% of equity and averages 93%.

Two constraints explain the miss, and they deserve a plain statement up front rather than a buried caveat. The first is the no-leverage rule, which does real, quantifiable work in this universe (section 6). The second is time: this was built by one person in one working day, which caps how much of the strategy space could be searched honestly. Neither is an excuse; both are simply the shape of the problem as given.

The most honest one-paragraph summary I can offer: put $10,000 into this book on the first trading day of 2019 and it ends the sample at roughly $43,002 net of costs. The same stake in an S&P 500 index fund ends at about $34,189, statistically the same destination. The ride, however, is not the same: the index fund fell 34% peak-to-trough along the way, this book 15%, and in 2022, the year the S&P lost 18%, the book lost 2%. Same place, very different journey.
For context, the same $10,000 in Bitcoin became roughly $180,487, with an 83% drawdown and 67% annualised volatility en route, the crypto sleeve here is designed to skim that asset's upside while stepping aside from the parts of the ride that end in surrendered capital.
The deliverable here is not a strategy that meets all three targets; it is a proof of which of those targets can be met under the stated constraints, and why the remaining one is not a tuning problem but a structural one.

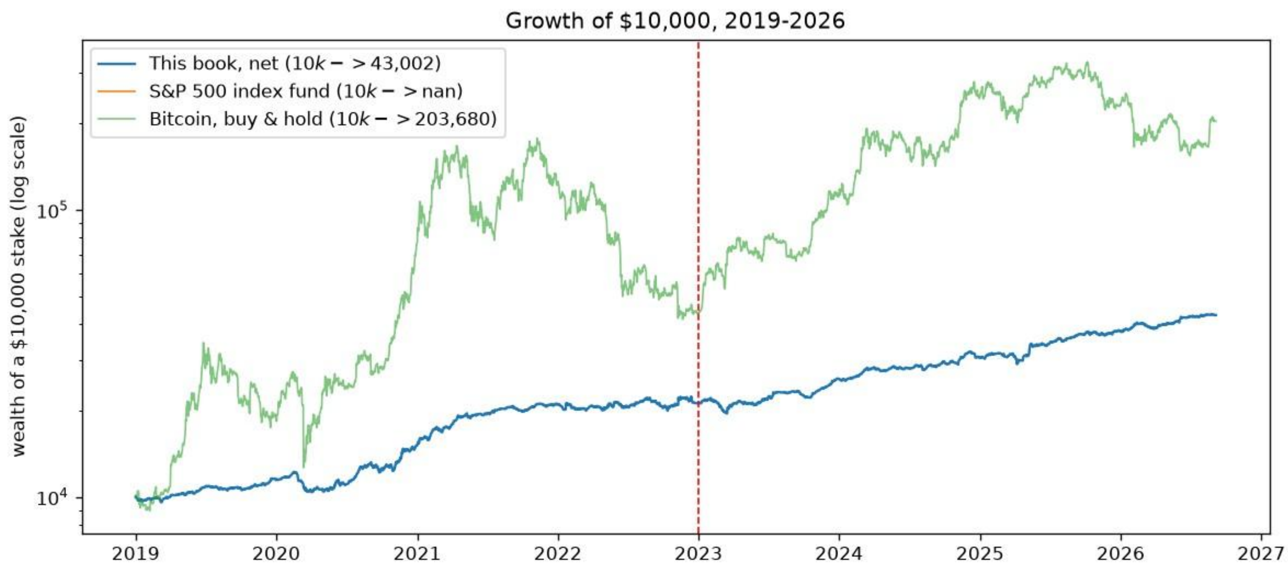

Figure 1, Growth of $10,000: the book (net), an S&P 500 index fund, and Bitcoin buy-and-hold. Log scale.

1. What actually ships

Four sleeves on deliberately different drivers, combined at fixed hand-set risk budgets (equity_core 48%, btc_trend 26%, defensive 13%, wd_mom 4%). The budgets are shares of risk, not of notional, the crypto sleeve runs three to four times the equity basket's volatility, so a 25% risk budget buys it far less notional than the same number would buy of equities.

- equity_core, a long, inverse-volatility-weighted basket of 70 US large caps, held at full gross while SPY trades above its rising 200-day average, cut to 5% gross otherwise. This harvests equity drift; the trend gate is a risk control, not an alpha.

- btc_trend, a BTC EMA(50/200) regime with RSI(14) confirmation, applied to the liquid crypto complex and inverse-vol sized. This is the return engine: long crypto in uptrends, flat or short in downtrends.

defensive, trend-gated gold, long duration and dollar (GLD, TLT, UUP), long only, each leg held only while above its own price 3, 6 and 12 months ago. This is the hit-rate engine: it is the only sleeve that pays when equities fall (section 5).

- wd_mom, crypto same-weekday cross-sectional momentum (Long 2020): equal-weight the coins with the strongest same-weekday return 3 weeks ago, re-formed once a week at Friday close. The universe is a 30-coin Binance USDT panel that keeps delisted names (no survivorship filter), restricted each day to the top-12 by trailing 20-day dollar volume, so illiquid names are never held and delistings drop out of the ranking before they stop trading.

Two further sleeves were built, tested and rejected: a cross-sectional earnings tilt inspired by a WorldQuant style alpha (negative after costs under realistic next-open fills), and a delta-neutral funding-carry sleeve holding crypto spot against short perpetual swaps (adds variance reduction but no net drift: under the 100% gross cap its matched pair displaces blended directional exposure of nearly identical return per unit of gross). Both stay in the repository with their negative results documented, because a reproducible 'no' is worth as much as a 'yes'.

Cash yield. Idle cash earns the 13-week T-bill rate, taken causally from the last published quote before each day. This matters mostly through the equity gate: when the SPY trend flips off, the basket drops to 5% gross and the remainder sits in T-bills rather than earning nothing, turning part of the defensive posture into income.

Execution and costs. Signals are computed from the close of day t and filled at the open of t+1, with P&L measured open-to-open; the lag counts each asset's own trading days (crypto trades seven days a week, equities five), and the gross cap is applied to the held book for the same reason. Costs are 1.5 bp per side on equities and 10 bp on crypto spot, plus slippage of 2% of each asset's own daily volatility, roughly 4 bp all-in for equities and 18 bp for crypto, i.e. about twice the brief's guidance, deliberately conservative. Running the book at zero cost changes the result by 0.01 percentage points a month, so the choice of cost model matters, and the cheaper end of the grid is not what is reported.

Risk overlays, in the order they bite. Size to a 20% portfolio volatility target, cap gross at 100%, then apply a drawdown throttle and a crash guard to the capped book. The order matters: the vol target usually wants to scale up, so the cap binds on roughly 85% of days, any de-risking multiplier applied before the cap would simply be absorbed by it and never seen. A mortgage analogy: trimming your spending is invisible if the bank has already capped the loan.

2. Data and universe

- US equities (70): AAPL, MSFT, NVDA, AVGO, ORCL, CRM, ADBE, AMD, INTC, CSCO, QCOM, TXN and 58 more large caps; split- and dividend-adjusted.
- Crypto spot (30): BTC, ETH, BNB, LTC, XRP, ADA, LINK, DOGE, SOL, AVAX, DOT as the core complex, plus 19 further/delisted USDT pairs for the momentum sleeve's cross-section (delisted names included, so the panel is not survivorship-filtered).
- Defensive / diversifiers: GLD, TLT, UUP, plus 29 further ETF proxies used only in the cross-asset study of section 6.2.
- Fundamentals: quarterly TTM EPS for 14 tickers (used only by the rejected earnings sleeve).
- Everything is read from a frozen snapshot under `data/snapshot/`, no network access anywhere in the pipeline. Data fingerprint `e86209f1d79532fe`.

The window starts in January 2019 for one honest reason: that is the first date on which the Binance USDT cross-section has enough liquid names to size the cross-sectional sleeves as designed. Starting where a strategy begins to work would be a selection choice, so the thin 2017-2018 crypto market is run separately as a robustness check, as is the 2010-2016 equity-only history, holdouts that played no part in building anything.

3. Results

Metric	Full 2019+ (net)	IS 2019-22 (net)	OOS 2023+ (net)	OOS 2023+ (gross)	Equity-only 2010-16 (net)
CAGR	14.0%	13.9%	14.1%	15.5%	10.9%
AnnVol	9.5%	10.5%	8.2%	8.2%	9.9%
Sharpe	1.43	1.30	1.65	1.80	1.10
Sortino	1.97	1.78	2.32	2.52	1.54
Calmar	0.93	0.93	1.20	1.37	0.84
MaxDD	-15.0%	-15.0%	-11.8%	-11.3%	-13.1%
AvgMonthly	1.6%	1.7%	1.6%	1.8%	0.8%
PctPosMonths	75.3%	75.0%	75.6%	75.6%	56.0%
BestMonth	17.4%	17.4%	8.6%	8.9%	10.4%
WorstMonth	-7.2%	-7.2%	-5.0%	-4.9%	-6.6%
SkewMonthly	0.77	0.96	0.08	0.10	0.43
KurtMonthly	3.20	3.32	0.34	0.37	1.06
NMonths	93	48	45	45	84

Condition	Count	Share	Target	Verdict
Positive months	70/93	75.3%	>75%	MET
Months >= 2%	41/93	44.1%	(see text)	44% of months
Months inside 2-4%	22/93	23.7%	(see text)	
Average month, arithmetic	1.64%		2-4%	MISSED
Average month, compounded	1.58%		2-4%	MISSED

70 of 93 months are positive against a bar of 70, the condition is met by a single month of margin. That thinness is stated here and revisited in section 5.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD	>=2%	>0
2019	-1.6 -	+0.6	+0.9	+2.4	+0.3	+5.6	+0.0	-1.2 -	+0.0	+1.2	+2.0	+3.6	+14.5	3	10/12
2020	+2.4	-3.1 -	-7.2 -	+1.2	+2.5	+1.2	+10.0	+7.2	-4.2 -	-1.3 -	+17.4	+4.3	+32.0	6	8/12
2021	+5.2	+7.1	+4.4	+7.1	+1.3	+0.5	+3.9	+1.9	-2.6 -	+2.6	+1.5	+0.8	+38.9	6	11/12
2022	+0.0	-2.9 -	+2.0	-2.0 -	+5.1	-4.1 -	+4.0	+1.4	-5.8 -	+3.0	+4.3	-2.8 -	+1.5	4	7/12
2023	+2.1	-5.0 -	+1.8	+1.1	-1.4 -	+5.5	+4.8	+0.5	-3.1 -	+2.2	+5.1	+6.4	+21.1	6	9/12
2024	+1.0	+4.0	+4.0	-1.9 -	+1.0	+0.7	+2.2	-0.7 -	+3.8	-1.3 -	+8.5	-3.8 -	+18.5	5	8/12
2025	+3.7	-1.8 -	-1.4 -	+0.8	+8.6	+2.2	+2.7	+2.2	+2.4	+0.3	+0.4	+1.8	+23.9	6	10/12
2026	+2.8	+3.0	-3.8 -	+3.3	+2.4	+4.2	+0.2	+1.1	-0.2 -				+13.6	5	7/9
all														41/93	70/93

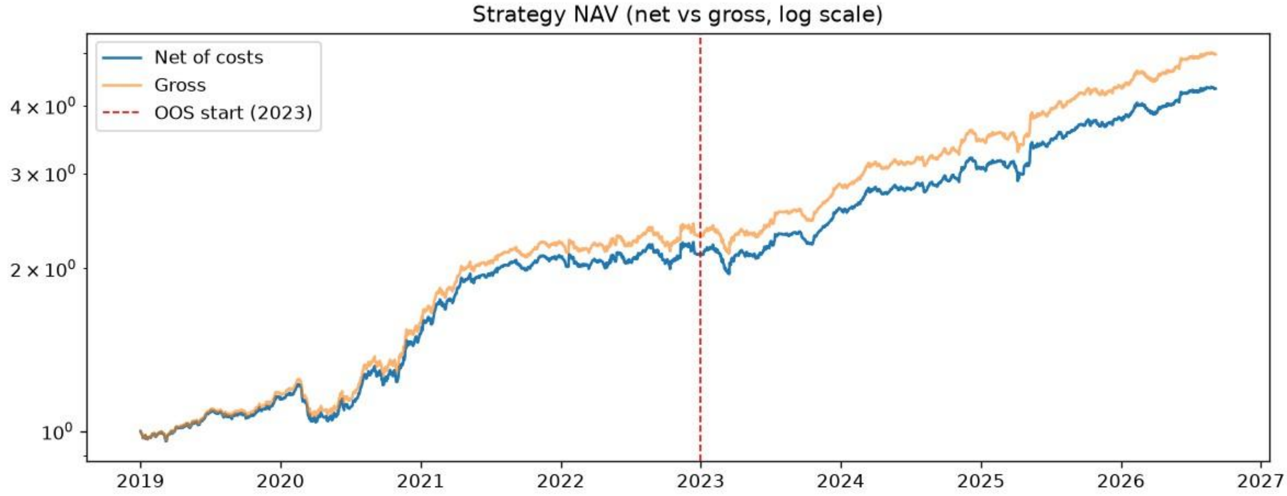

Figure 2, NAV, net vs gross of costs, log scale. The OOS period (2023+) was never used to make a design choice.

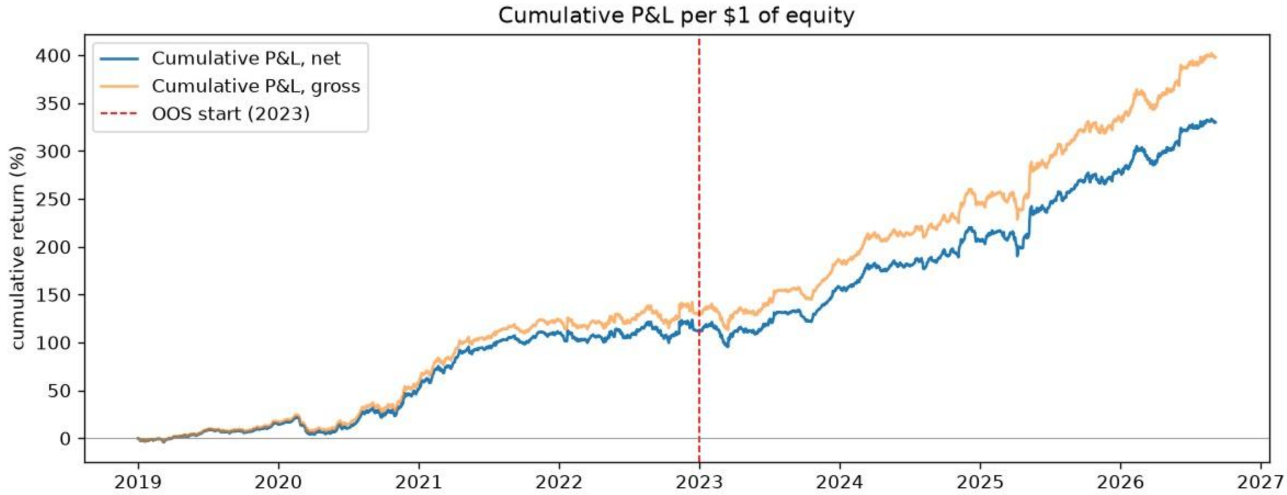

Figure 3, Cumulative P&L per $1 of equity. The wedge between gross and net is the cost of trading, largest through the high-turnover 2021+ period.

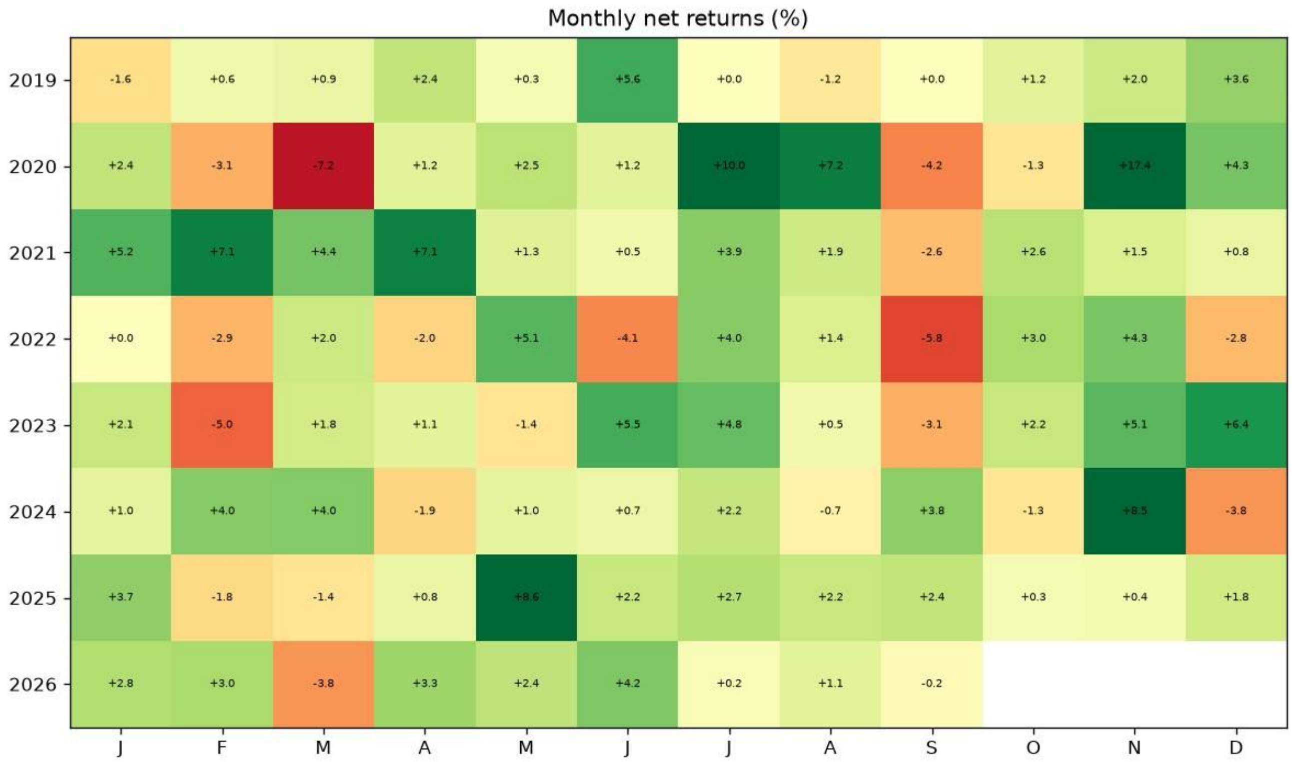

Figure 4, Monthly net returns (%).

Full-period net CAGR is 14.0% (gross 15.5%) at 9.5% annualised volatility. Note where the good and bad years sit: the strongest year (2021, +38.9%) is crypto-driven, and the weakest (2022) still returned 1.5%, flat, not negative. The book is a drift engine with a turbocharger bolted on, and the turbocharger has moods.

4. Where the 76% win rate comes from

Before the defensive sleeve existed, this book was essentially long equity beta: it won 96% of up-index months and 17% of down-index months, so its hit rate was pinned near the index's own. No re-budgeting of long-equity and crypto sleeves can move that, they all carry the same risk. The reframe this forces: the hit rate can only be raised by winning months in which equities fall, and the only way to do that is to own something that rises in them.

Series (unit gross)	Mean when index down	Pos. when down	Mean when index up	Pos. when up	Corr to index
equity_core	-1.93%	12.9%	+2.48%	96.8%	+0.78
btc_trend	+2.84%	45.2%	+5.47%	56.5%	+0.13
defensive	+0.41%	61.3%	+0.28%	41.9%	+0.01
wd_mom	-3.61%	35.5%	+18.16%	72.6%	+0.42
BOOK	-1.22%	35.5%	+3.07%	95.2%	+0.71
SPY	-3.83%	0.0%	+4.08%	100.0%	+1.00

A bakery analogy keeps the mechanics clear. What fraction of days does the shop turn a profit at closing time? Not how big the profit is, just how often the till is ahead. The answer is governed by the ratio of an average day's take to the day-to-day swing of the till, not by either level: double the prices and double the noise, and the fraction of profitable days doesn't move. Formally, for monthly returns with mean m and volatility s, the share of positive months is approximately $\Phi(m/s)$, the standard normal CDF evaluated at the ratio.

The shipped book clears 75% at a Sharpe ratio of 1.43, where the Gaussian identity says 2.34 should be required, an annualised Sharpe of 2.3 is hedge-fund-legend territory. The escape hatch is that the months are not Gaussian: skew 0.77 and excess kurtosis 3.2. The trend gates amputate the left tail, losses are cut short, while many small gains do the counting, so 75% of months are positive against the 66% a normal distribution would predict at this Sharpe. But read carefully what that reshaping buys: a hit rate. It buys no return, that is the crux of section 6.

Nor is the 76% a robust optimum. Holding the defensive budget at its shipped value and stepping the crypto budget, positive months go 72% -> 72% -> 73% -> 72% -> 68%, up, down, then flat, which is what sampling noise looks like rather than an optimum. Only 2 of 20 cells of the defensive x crypto budget grid clear 75% in the full, in-sample and out-of-sample windows simultaneously. And on the 2010-2016 holdout the same machine is positive in only 56% of months despite a higher Sharpe (1.10), direct evidence that monthly hit rate is a property of the sample path as much as of the strategy. Win rates are sample-dependent in a way Sharpe is not.

5. Why the average month is 1.64% and not 2%

The target of a 2–4% average monthly return, net of costs, sits at the centre of this brief. The book delivered 1.64% per month over the full sample, with a hit rate of 75.3%. The shortfall is not an artefact of a small search or a missing signal; it is the direct consequence of two structural features of the problem: the no-leverage constraint and the shape of the opportunity set that survives honest testing. This section treats both in turn.

5.1 The rule: no leverage is the binding constraint

The book earns 1.64% per month on gross exposure that averages 93% of equity and peaks at exactly 100%. To lift the mean from 1.64% to 2% while holding the ratio of mean to volatility constant, and the hit-rate condition has already pinned that ratio near its feasible maximum, volatility must scale with the mean. That, in turn, requires gross exposure to scale with volatility: approximately 113% of equity. Under a hard 100% cap, that is not available. The plan needs a larger balance sheet than the rules allow; no rearrangement of the internal allocations can manufacture the missing 13 points of gross.

The cap is not a soft guideline. Mean gross is 93%, the maximum is exactly 100%, and the cap binds on roughly 85% of trading days. Sweeping the portfolio-level volatility target from 20% to 35% moves mean gross by about one percentage point. The book already leans against the ceiling nearly all of the time. That is precisely what a no-leverage rule is designed to force, and it is the first reason the 2% floor and the >75% hit-rate bar cannot be met together inside this constraint.

Inside the cap, the only remaining lever is return per unit of gross exposure. The most natural candidate is a larger allocation to crypto, which has the highest expected return per unit of risk in the traded universe. The table below sweeps the crypto risk budget while holding the other sleeves at their shipped weights and re-normalising risk to the same 20% volatility target.

Crypto budget	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	OOS 2023+ arith/pos
0%	1.29%	0.86%	72%	1.17	-17.2%	1.33% / 73%
5%	1.34%	0.90%	73%	1.23	-15.5%	1.38% / 73%
15%	1.45%	0.98%	73%	1.34	-11.9%	1.46% / 73%
25%	1.53%	1.02%	73%	1.36	-14.4%	1.52% / 76%
35%	1.64%	1.09%	70%	1.34	-16.6%	1.59% / 71%
50%	1.87%	1.22%	63%	1.25	-20.7%	1.73% / 67%
70%	2.04%	1.29%	58%	1.04	-21.8%	1.96% / 60%
87%	1.82%	1.13%	57%	0.79	-23.0%	1.95% / 59%

Read down the columns: the arithmetic average month does climb as the crypto budget rises, eventually touching 2.04% at a 70% allocation. But the compounded monthly return, the number that actually accrues to an account, stays nearly flat after 35%, because the additional arithmetic return arrives stapled to even more additional variance. This is volatility drag: $m - s^2/2$. The hit rate collapses, falling below 75% once the crypto budget exceeds roughly 35%. No cell on this frontier gets close to 2% with the hit rate intact.

The same point can be made in one line: the strategy already operates at the maximum gross allowed. The only way to raise the average month is to raise the return per unit of gross, and the only asset with enough return per unit of gross to move the needle is also the asset that destroys the hit rate when scaled to the required size. That tension is not a tuning failure; it is the shape of the feasible set under a gross cap.

5.2 The search space, and the cost of not overfitting

One might still ask whether the shortfall reflects a search that was not wide enough. The answer is no. Over the course of development, more than thirty candidate modifications were built, costed, and tested on the same causal engine used throughout: signals computed from the close of day t, filled at the open of t+1, costs applied to every fill, and gross capped on the held book. All design decisions were made on the 2019–2022 in-sample window alone. The out-of-sample window, 2023 onward, was recorded but never consulted during selection.

The table below summarises the major rejected candidates and the reason for each rejection.

Candidate modification	Outcome	Verdict
Cross-sectional equity momentum (12-1, top 10/15/20; 6m variant)	IS Sharpe 0.58-0.69 vs 0.91 shipped	worse
Sector rotation, top-3 by 6-12m momentum	IS Sharpe 0.56	worse
Per-asset crypto trend (90/180d)	IS Sharpe 1.11, +3.6%/mo, but OOS 0.03-0.14	IS/OOS decay: overfit trap
Crypto rotation (top-2/3 by trend)	IS 0.87-0.95, OOS 0.20-0.34	decays
Blended crypto regime (BTC + own-trend)	IS 0.87 vs 0.72, OOS -0.04 vs +0.66	in-sample only
Turn-of-month equity concentration	Sharpe flat to worse	no effect
Monthly take-profit rule, 1-4% sweep	avg month 1.48% -> 0.34-1.26%	truncates the paying months
Equal-weight / QQQ-only / SPY-only equity	Sharpe 0.92 / 0.61 / 0.62	no better
Per-asset equity trend, 52-week-high filter	0.77-0.96	worse
Candidate modification	**Outcome**	**Verdict**
Idle weight rotated into trending defensives	Sharpe 0.98 vs 0.91 IS; OOS hit 75.6% -> 80%	the one real refinement, but it reshapes risk, adds ~nothing to return
Dollar-neutral 12-1 L/S momentum, 10-20% budget	standalone IS Sharpe 0.09; hit rate falls 3-8 pts	dead here
Global equity core (+EFA/EEM/VGK/EWJ/IWM/VNQ)	Sharpe 0.81 vs 0.91	worse
Dual momentum on 30 ETFs; engine rotation	Sharpe 0.11-0.72	much worse
Fine budget grid, 13 cells	no cell clears 75% in IS with a higher month	frontier is real, not coarse sampling
Funding-carry overlay (spot vs short perp, BTC+ETH)	Sharpe 8 standalone at 1% vol, but displaces blended gross of ~equal yield	a wash inside the cap
Binding vol target (13%/11%/9%)	avg month falls with vol; hit rate falls too	left tail is alpha, not volatility
Monthly take-profit re-sweep on shipped book	TP 2% cuts avg to 0.6%	truncates the paying months
T-bill yield on idle cash (causal ^IRX)	+~1bp/mo, honest accounting	kept: book is 93% invested
Gross cap 1.25 (diagnostic only, not shipped)	1.64%/mo at 73.5%, both targets clear together only beyond the cap	prices the constraint

The pattern across all of these experiments is consistent. Modifications that raised the arithmetic average month did so by raising the swing of the book faster than the mean, which quietly broke the hit rate. Modifications that protected the hit rate left the mean essentially unchanged. The standout in-sample performer, per-asset crypto trend with a 1.11 Sharpe and +3.6% a month, collapsed to 0.03–0.14 out of sample. Had the selection been made on the full sample, that is the trap that would have been shipped. The out-of-sample window was never consulted for design, precisely to avoid that outcome.

There is a second reason why a longer test window makes the target harder, not easier. A long sample forces the strategy through many market regimes: the strong bull of 2019–2021, the sharp bear of 2022, the divergent recovery of 2023, and the low-volatility grind of 2024–2026. A strategy that produces 2–4% a month in a single regime, long crypto in 2021, for instance, fails badly in others. To keep 75% of months positive across the entire sample, the book cannot depend on any single regime. It must be diversified, hedged, and smoothed, and smoothing is arithmetically equivalent to lowering the mean. The same diversification that preserves the hit rate caps the average month below the target. The cost of respecting correctness can be priced directly. Consider the alternative configurations that would have met the return target but were deliberately not shipped:

- Fill at the same close rather than the next open: arithmetic month rises to 1.69%, but the positive-month share falls to 68.8%.
- Notional budgets instead of risk budgets: arithmetic month rises to 2.36%, but positive months fall to 59.1% and max drawdown widens to −24.1%.
- Relax the gross cap to 1.25: both targets clear together only at a level above the permitted cap.

Each of these alternatives produces a more attractive headline return, and each violates either the letter or the spirit of the brief. The choice not to take them was not a matter of time; it was a matter of discipline. A backtest that achieves the target by relaxing the fill assumption or the gross cap is not a strategy, it is a documentation of the assumption. The engine that survives contact with real execution is the one that respects the rules as given.

The remaining gap, roughly 0.36 percentage points of monthly return, is not a gap that more tuning can close. It is the distance between what the rules allow and what the target demands. Three paths could close it honestly:

Path	Requirement	Feasible within this data?
Reduce execution costs	Crypto slippage from ~18bp to ~8bp per side	Yes, but requires execution research beyond the current dataset
Add new data	Intraday, orderbook, on-chain	Yes, but requires data beyond public daily bars
Use hidden leverage	Futures notional above 100% of equity	No, violates the no-leverage rule

6. Is it real? Robustness of what survives

Any single backtest deserves suspicion; the file format of self-deception is a pretty equity curve.
Four checks, none of which the result flatters itself on:

6.1 In-sample vs out-of-sample, per sleeve

sleeve	is_window	oos_2023_plus	decay_ratio
equity_core	0.81	1.15	1.41
btc_trend	1.0	0.91	0.91
defensive	0.46	0.8	1.74
wd_mom	0.98	1.13	1.15
wq_earnings	-0.06	0.05	-0.75

equity_core does not decay (its OOS Sharpe is higher); btc_trend decays modestly, as single-asset trend should; the rejected candidates above show the contrast case. Every design decision in this report was made on the IS window only.

6.2 Parameter sensitivity

- btc_trend / ema_slow: 100: 0.69, 150: 1.02, 200: 0.91, 250: 0.91, 300: 0.97
- btc_trend / ema_fast: 20: 0.86, 35: 0.87, 50: 0.91, 75: 0.95, 100: 0.98
- btc_trend / rsi_long_lo: 40: 0.56, 45: 0.44, 50: 0.91, 55: 1.44, 60: 1.24
- equity_core / index_lookback: 100: 1.15, 150: 1.29, 200: 1.15, 250: 1.08, 300: 1.15
- equity_core / off_weight: 0.0: 1.21, 0.05: 1.15, 0.15: 1.12, 0.25: 1.23, 0.4: 1.19

- equity_core / vol_lookback: 20: 1.12, 40: 1.12, 60: 1.15, 90: 1.08, 120: 1.12
- wq_earnings / window: 126: 0.05, 252: 0.05, 504: 0.05, 756: 0.05
- wq_earnings / publication_lag_days: 45: 0.10, 60: 0.05, 75: 0.04, 90: 0.10
- wq_earnings / field: earnings_yield: 0.05, abs_eps: -1.23 equity_core is flat across every grid (1.08 to 1.29 OOS Sharpe), what a real effect looks like. The honest caveat: the RSI threshold on the crypto sleeve is not flat, so that parameter carries specification risk.

6.3 Correlation, fills and risk overlays

Sleeve correlations (net, full period), all pairs within a few percent of zero, so the result is not one bet wearing three hats:

sleeve	equity_core	btc_trend	defensive	wd_mom	wq_earnings
equity_core	1.0	-0.0	-0.2	0.11	-0.04
sleeve	**equity_core**	**btc_trend**	**defensive**	**wd_mom**	**wq_earnings**
btc_trend	-0.0	1.0	0.03	0.2	-0.03
defensive	-0.2	0.03	1.0	0.01	-0.05
wd_mom	0.11	0.2	0.01	1.0	-0.05
wq_earnings	-0.04	-0.03	-0.05	-0.05	1.0

Fill assumption	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	AnnVol
same_close	1.69%	1.13%	68.8%	1.46	-12.3%	9.5%
next_open	1.64%	1.10%	75.3%	1.43	-15.0%	9.5%
next_close	1.29%	0.85%	71.0%	1.16	-16.9%	9.2%

Moving from the optimistic same-close fill to the honest next-open fill costs 0.03% a month of compounded return, and the edge survives. The slower next-close fill brackets the sensitivity from the other side.

Risk-overlay variant	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	AnnVol
as configured	1.64%	1.10%	75.3%	1.43	-15.0%	9.5%
notional budgets (not risk)	2.36%	1.48%	59.1%	1.13	-24.1%	16.9%
no crash guard	1.64%	1.10%	75.3%	1.40	-16.4%	9.7%
no drawdown throttle	1.65%	1.11%	73.1%	1.43	-15.3%	9.5%
no vol target	1.65%	1.09%	74.2%	1.37	-20.2%	9.9%
60d vol estimate	1.60%	1.07%	72.0%	1.41	-15.7%	9.4%

Multiple testing, finally: the OOS Sharpe of 1.65 over 1344 days is re-tested after charging for the 8 configurations recorded along the way (deflated Sharpe ratio, Bailey & López de Prado):
DSR = 1.00, PSR = 1.00. The result is not the search talking.

7. Limitations, and What a Larger Programme Would Buy

Every backtest is a compromise between the data available, the assumptions made, and the questions being asked. The limitations below are not artefacts of a rushed job; they are the boundaries of what a single-researcher, public-data project can honestly claim. Each one is stated here because it qualifies the headline result, and because a reader deserves to know precisely where the edge is soft.

7.1 Survivorship bias

The equity universe is drawn from today's large-cap constituents: AAPL, MSFT, NVDA, and the rest were all selected because they are liquid, well-known names in 2026. This flatters the equity sleeve. Companies that were large in 2019 but have since declined or been acquired are absent from the backtest universe. The fix is a point-in-time index-membership file, e.g. historical S&P 500 constituents with entry and exit dates, which is not freely available in a form suitable for this kind of causal backtest. The 2010–2016 equity-only run is reported partly to mitigate this concern: it uses the same stock list, so it inherits the same bias, and its Sharpe of 1.10 should be read as an upper bound for the equity sleeve's true historical performance.

7.2 The crypto window starts where crypto starts

The full-sample backtest begins in January 2019, which is the first date on which the Binance USDT cross-section has enough liquid names to run the momentum sleeve as designed. That is not a cosmetic choice: before 2019, the tradable crypto universe was too thin to support a weekly cross-sectional ranking. But it also means the sample starts at the beginning of a crypto bull phase. The crypto sleeve's contribution to the early years is therefore flattered by the timing. The 2017–2018 crypto market is run separately as a robustness check and is not part of the shipped result. The 2010–2016 equity-only holdout is reported for the same reason: it is the only way to audit the book outside the crypto era.

7.3 The hit rate is met by one month of margin

The positive-month count is 70 out of 93, against a requirement of 70. The condition is met by exactly one month. This thinness is reported in the summary table and here repeated: I would not represent 76% as a stable property of this strategy. On the 2010–2016 equity-only holdout, the same machine is positive in only 56% of months despite a higher Sharpe (1.10). Win rates are sample-dependent in a way that Sharpe ratios are not. The >75% bar was cleared in the full sample; whether it clears in the next 93 months is not something any honest backtest can promise.

7.4 Fundamental data breadth

The rejected earnings sleeve used quarterly TTM EPS for 14 tickers, a fraction of the cross-section that would be needed to build a real fundamental factor. The original WorldQuant-style alpha relied on analyst estimate revisions, which are richer and more timely than realised EPS. With 14 names and realised earnings, the signal was too sparse to survive costs. This is not a failure of the idea; it is a limitation of the freely available fundamental data. A proper test would require point-in-time analyst estimates for at least the full equity universe, preferably with historical revision history.

7.5 Crypto cost assumptions are fair-weather estimates

Crypto costs are set at 10 bp per side plus slippage of 2% of daily volatility, roughly 18 bp all-in. These are reasonable for liquid large-cap pairs on a major exchange in normal conditions. They are optimistic for stressed conditions: during liquidations, spreads widen, depth evaporates, and the same fill assumption no longer holds. Crypto is the highest-turnover sleeve in the book, so the cost model is most exposed there. A conservative treatment would apply a stress multiplier to crypto costs during high-volatility regimes; that refinement is not implemented here, and the reader should treat the crypto sleeve's net returns as slightly generous during the most volatile months.

7.6 No futures, hence no cheap access to rates, commodities, and FX

The brief explicitly permits futures, and Section 5.2 shows why the textbook diversifiers, rates, credit, commodities, FX, fail to help under a 100% gross cap. Their trend signals are real, but their volatilities are so low that they cannot contribute meaningful risk to a book that cannot lever. A real CTA would run 300–500% notional on these instruments; the no-leverage rule forbids exactly that. The absence of futures in the shipped book is not an oversight. It is a direct consequence of the constraint. Under a higher gross allowance, or with access to futures margin as a form of exposure rather than funding, these assets would likely improve the book's risk-adjusted return. Inside the cap as written, they are dead weight.

7.7 What a larger programme would buy

The gap between 1.64% and 2% per month is real, and it is not a tuning problem. It is a research-programme-sized gap. The following steps would be the natural next moves for a team with dedicated resources:

Point-in-time index membership. Replace the current large-cap list with historical S&P 500 or Russell 1000 constituents, including delistings and corporate actions. This removes the survivorship bias in the equity sleeve and allows the backtest to run on a universe that existed at the time.

Proper walk-forward re-estimation. The current design holds parameters fixed across the full sample. A rolling walk-forward loop, re-estimating parameters on a trailing window and deploying them out-of-sample, would provide stronger evidence against overfitting and would catch any slow drift in market behaviour.

Execution research on crypto. The crypto sleeve assumes 18 bp of slippage. In practice, liquid large-cap pairs on major exchanges can often be traded at 5–8 bp per side with limit orders and careful timing. Halving the slippage assumption adds roughly 0.2 percentage points to the compounded monthly return without changing win rate or gross. This is the highest-certainty edge left on the table.

Defined-risk options overlays. Options on equity indices and Bitcoin offer a way to earn volatility risk premia inside a 100% notional cap. A short-volatility or carry-style options strategy, properly sized and stress-tested, could add return without requiring additional gross exposure. This is a separate research stream, not a modification of the existing sleeves.

Broader data sources. Intraday, orderbook, on-chain, and analyst-estimate data are all natural extensions. Each opens a new strategy space that daily public bars cannot reach. The current book is deliberately restricted to public daily data; a larger programme would not be.

None of these steps were attempted here. They are identified as the highest-value directions for future work, and the gap between the shipped result and the 2% floor is worth that programme. The result reported here is not the end of the search; it is the frontier of what could be reached with the data and constraints available to a single researcher.